ASSIGNMENT - 2

Fundamentals of Cyber Security (TCS-495)

Name - Priyanshi Rawat
Sec - B
Roll No - 47
Uni Roll No - 2025390
St ID → 240222161

Git Link → /priyanshi-r76/ CyberSecurityAssignmat/

Ans:1 Computer networking is one of the most important foundations of ethical hacking and cyber security. Ethical hackers would work with networks to identify vulnerabilities, monitor traffic, detect attacks and secure communication systems. Almost every cyberattack involves networks in some form, therefore understanding networking is essential for any cyber security student.

Networking helps ethical hackers understand how devices communicate with each other over the internet or local networks. It allows security professionals to analyze data packets identify suspicious traffic, secure servers, and prevent unauthorized access

TCP/IP

1. understanding Network Communication:
   TCP/IP explains how data travels b/w devices through packets. Ethical hackers need this knowledge to analyze attacks and communication patterns.
2. Packet Analysis
   Tool such as wireshark captures packets based on TCP/IP protocols. Understanding TCP/IP helps in analyzing packet content and identifying malicious traffic.
3. Vulnerability Detection.
   Many attack target TCP/IP weaknesses such as spoofing, session hijacking, and SYN flooding
4. Network Troubleshoting
   Cyber security professionals use TCP/IP concepts to troubleshoot connectivity problems and detect suspicious activities.
5. Penetration Testing
   Ethical hacking use TCP/IP knowledge for scanning ports, identifying open services, and testing networks security.

Ans:2

	NAT	Bridged	Host-Only
Meaning	V/M shares host's IP address using Network Address Translation.	VM acts like separate device on physical network	VM communicates only with host system.
Internet Access	Yes	Yes	No
Communication with host	Yes	Yes	Yes.

Communication with other Devices	Limited	Yes	No
Real Use Case.	Safely browsing internet inside VM	Network penetration testing lab	Malware testing in isolated environment.

Ans 3 The TCP 3-way handshake is the process used to establish a reliable connection b/w a client & server before data transmission begins.

Steps of TCP 3-way Handshake :-

Step 1 - SYN
The client sends a SYN (synchronize) packet to server requesting connection establishment.

Step 2 - SYN-ACK
The server replies with SYN-ACK (Synchronize Acknowledge) packet confirming the request.

Step 3 - ACK
The client sends an ACK (Acknowledgement) packet to complete the connection.

Client $\xrightarrow{SYN}$ Server

Client $\xleftarrow{SYN\text{-}ACK}$ Server

Client $\xrightarrow{ACK}$ Server

Connection Established.

Risks if Handshake Security is Weak :-

1. SYN Flood Attack Attackers send multiple SYN requests w/o completing connections, causing server overload.
2. Session Hijacking Attackers may steal active sessions & impersonate users.
3. Spoofing Attack Fake IP addresses can be used to exploit systems.
4. Denial of Service (DoS) Weak handshake security may lead to service disruption.

Ans 4

IPv4	IPv6
1. 32-bit address	1. 128-bit address
2. Uses decimal format	2. Uses hexadecimal format
3. Example: 192.168.1.1	3. Example: 2001:db8::1
4. Limited address space.	4. Huge address space.
5. Supports around 4.3 billion addresses	5. Supports extremely large number of addresses.

IP Address	MAC Address
1. logical Address	1. Physical hardware address
2. Assigned by network	2. Assigned by manufacturer
3. Can change	3. Usually permanent
4. Used for routing	4. Used for device identificati-

DNS	Ping
1. Convert domain names into IP addresses	1. Tests connectivity b/w devices
2. Uses DNS protocol	2. Uses ICMP protocol
3. Ex: google.com → IP	3. Check response time & reachability.

Ans 5: Subnetting Problem

Given Network: 192.168.10.0/24

Requirement: 4 departments with equal hosts.

Step 1 → Determine Borrowed Bits.

To create 4 subnet ⇒ $2^2 = 4$

therefore 2 borrow bits.

Original subnet mask: /24

New subnet mask: /26.

a) New Subnet Mask

255.255.255.192

b) Number of subnet Created

4 subnets

c) Number of Usable Hosts Per Subnet

Formula:

$= 2^{(32-26)} - 2$

$= 2^6 - 2$

$= 64 - 2$

$= 62$ usable hosts

d) First Four Subnet Addresses

Subnet Adr	Network Adr
1	192.168.10.0/26
2	192.168.10.64/26
3	192.168.10.128/26
4	192.168.10.192/26

Ans. 6.

IP Address Calculation.

Given: 172.16.25.130/26

Step 1 - Understand Subnet

/26 mean

255.255.255.192

Block size

256 - 192 = 64.

Subnet range : 0-63
64-127
128-191
192 -255

IP 172.16.25.130 falls inside: 128-191 range.

a) Network range

172.16.25.128

b) Broadcast Address

172.162.25.191

c) Usable Host Range

172.16.25.129 to 172.16.25.190

d) Total Usable Host

$2^{(36-26)} - 2$

$2^6 - 2$

$64 - 2$

62 usable host.

Q1 Virtualization Practical

1) Installation of VMware Workstation.

~~Downlo~~

1. Download VMware workstation.
2. Run setup file.
3. Accept license agreement
4. Complete installation process
5. Launch VMware Workstation

2) VM ware Creation steps.

1. Click "Creat a New VM"
2. Select "Typical Installation"
3. Choose "I will install ~~op~~ OS later"
4. Select Linux OS
5. Select Ubuntu 64-bit version
6. Enter VM name.
7. Allocate storage size.
8. Customize hardware settings.

RAM → 2048MB

Storage → 20GB

Q2. Wireshark Packet Capture

Wireshark is a network packet analyzer used to capture & inspect network traffic in real time.

It helps cyber security professionals analyze packets, troubleshoot networks detect suspicious traffic, and monitor communication b/w devices.

Practical Observation.

a) Source IP

2405:201:6811:e082:5c91:e7f7:4bcf:2205

b) Destination IP:

2404:6800:4000:1016::65

c) Protocol Used:

ICMPv6

d) Packet Count

22 packets

Q9 → Student Registration Form → created → github

How Poor Validation leads to XSS Attack?

Ans. It allows attackers to inject malicious scripts into i/p files.

If these scripts are executed inside another user's browser, the attack is called Cross-Site Scripting (XSS)

XSS attacks can:

- steal cookies
- hijack session
- redirect user to malicious website
- steal sensitive info

Proper validation and sanitization prevents XSS attack.

Q10 → PHP prog → written → github

a) SQL injection is where attacker inserts malicious SQL command into user i/p fields

Attacker may:
- bypass authentication
- steal database info
- modify records
- delete data

b) Prepared statements seprate SQL commands from user i/p. User i/p is treated only as data and not executable SQL codes, preventing attacker from injecting malicious queries

c)

Phishing	Spear Phishing
1. Generic attack targeting many users.	1. Targeted attack against specific person.
2. Uses fake email & website	2. Uses personalized info
3. Broad attack	3. Focused attack
4. Lower success rate	4. Higher success rate

Bonus question → completed → github.